SESSION ONE

PROBLEM ONE
Byron loves playing electronic games. The new Playmaster Game Console with the three new games he wants to play will cost \$1032. To pay for it he will have to take out a personal loan.

His Credit Union offers him a personal loan which charges interest at 12 % per annum, compounding monthly. The loan is intended to be repaid with 6 equal monthly payments of \$180.00.

(a) What is the monthly interest rate, written as a decimal correct to two decimal places ?

$$\textit{Monthly Interest rate :} = \frac{12\%}{12} = 1.0\,\% = 0.01$$

(b) Using B_n to represent the balance of the loan after *n* months, write down a recurrence relation to model this loan situation.

$B_0 = \$1032,\ B_{n+1} = 1.01B_n - 180$

The amortisation table for Byron's loan is shown below.

Payment number	Payment made	Interest paid	Principal reduction	Balance of loan
0	0.00	0.00	0.00	1032.00
1	180.00		169.68	862.32
2	180.00	8.62		690.94
3	180.00	6.91	173.09	
4	180.00	5.18	174.82	343.03
5	180.00	3.43	176.57	166.46
6				0.00
Total				

(c) Calculate, correct to the nearest cent :

(i) the interest that will be paid with the first payment

Interest $= \$1032.00 \times 0.01 = \$10.32$

(ii) the amount that is reduced from the principal with the second payment

Principal reduction $=$ *payment – interest* $= \$180.00 - \$8.62 = \$171.38$

(iii) the balance of the loan after the third payment is made.

Balance of loan $=$ *previous balance – principal reduction*
$= \$690.94 - \$173.09 = \$517.85$

Looking again at this partially completed amortisation table, we see that the balance owing after the fifth payment is quite a lot less than \$180, meaning that the final payment to amortise this loan will be less than \$180.

(d) Calculate the amount of interest to be charged for the final payment, correct to the nearest cent.

Interest payable for the last payment $= 166.46 \times 0.01 = \$1.66$

(e) Hence determine the amount of the final payment, correct to the nearest cent.

Final payment $= \$166.46 + \$1.66 = \$168.12$

(f) How much interest in total, correct to the nearest cent, does Byron pay on his loan?

Total Interest $= 10.32 + 8.62 + 6.91 + 5.18 + 3.43 + 1.66 = \$36.12$

Byron's friend, Garry, has also bought the Playmaster Game Console with three special games for \$1032. He paid for it using a personal loan from his bank, which charged 8.45 % per annum compounded monthly. However, under the bank's terms and conditions for a personal loan, Garry is required to make twelve monthly payments of \$90.00 to amortise this loan.

(g) Calculate the total amount of interest paid by Garry on his loan, correct to the nearest cent.

Interest $=$ *total paid – amount borrowed* $= 12 \times 90 - 1032 = \$48.00$

A major retailer of electronic games has made financial arrangements with a finance company to offer a special deal. The Playmaster Games Console with two special games is offered for \$998 cash or \$132.35 per month for eight months on a reducing balance loan.

(h) What interest rate, correct to one decimal place, is being charged by the finance company ?

Show details of any Finance Solver calculations in the table provided below.

N	**I(%)**	**PV**	**Pmt**	**FV**	**PpY, CpY**
8	*0*	*998*	*–132.35*	*0*	*12*

Solve for ***I%*** *: 15.99866…*
The interest rate is 16.0 % per annum.

SESSION TWO

PROBLEM ONE
Nellie is planning to save money for her "Schoolies" activity at the end of the year. She has an account with \$1500 in it already, and each month she will add another \$80 from her part-time job. The account pays 4.2 % p.a. interest, compounded monthly.

(a) What is the interest rate per month?

Monthly Interest rate : $= \dfrac{4.2\%}{12} = 0.35\ \% = 0.0035$

(b) Using B_n to represent the balance of the account after n months, write down a recurrence relation to model this investment situation.

$B_0 = \$1500 \quad B_{n+1} = 1.0035B_n + 80$

(c) Use your calculator to determine recursively the value of the investment after Nellie has made four payments.

Payment number	Calculation details	Value of investment
0		1500.00
1	*1500 × 1.0035 + 80*	*1585.25*
2	*1585.25 × 1.0035 + 80*	*(1670.7983…) 1670.80*
3	*1670.7983… × 1.0035 + 80*	*(1756.6461…) 1756.65*
4	*1756.6461… × 1.0035 + 80*	*(1842.7944…) 1842.79*

The value of the investment after four payments is \$1822.69.

(d) Nellie will close the account after ten (10) months. How much money, correct to the nearest cent, will she have for her "Schoolies" activities ?

Show details of any Finance Solver calculations in the table provided below.

N	**I(%)**	**PV**	**Pmt**	**FV**	**PpY, CpY**
10	*4.2*	*– 1500*	*– 80*	*0*	*12*

Solve for ***FV*** *: 2366.0529…*
The balance when the account is closed is \$2366.05.

PROBLEM TWO

Nellie's Aunt Selma is about to retire and is wondering what to do with her superannuation money.

One option is to invest her \$550 000 in an annuity paying 4.6 % per annum compounded monthly.

(a) How much would she receive per month if the annuity is for 20 years?

Show details of any Finance Solver calculations in the table provided below.

N	I(%)	PV	Pmt	FV	PpY, CpY
240	*4.6*	*– 550 000*	*0*	*0*	*12*

Solve for ***Pmt*** *: 3509.3302…*
The monthly payment would be \$3509.33

(b) Selma thought that this monthly payment did not support her intended lifestyle, and wanted to receive more. How long, correct to the nearest month, will this \$550 000 annuity invested at 4.6% compound interest p.a. compounded monthly last, if she is paid \$3750 per month?

Show details of any Finance Solver calculations in the table provided below.

N	I(%)	PV	Pmt	FV	PpY, CpY
0	*4.6*	*– 550 000*	*3750*	*0*	*12*

Solve for ***N*** *: 215.902…*
The payments would last for 216 months (18 years exactly)

(c) Her accountant has suggested that she check other annuity providers to find if one can provide a better interest rate. What interest rate, correct to two decimal places, would allow Selma to receive \$3750 per month for 20 years, based on her \$550 000?

Show details of any Finance Solver calculations in the table provided below.

N	I(%)	PV	Pmt	FV	PpY, CpY
240	*0*	*– 550 000*	*3750*	*0*	*12*

Solve for ***I%*** *: 5.3922…*
The required interest rate would be 5.39 % p.a.

Selma is in good health and believes that she will live longer than another 20 years, so she looked at investing in a perpetuity.

(d) If the interest offered is 5.4 % per annum, compounded monthly, how much will Selma receive per month, correct to the nearest cent ?

$$\textit{Payment (interest)} = \frac{5.4 \times 550000}{100 \times 12} = \$2475.00$$

(e) If Selma wanted to receive **at least** \$2750 per month from this perpetuity, how much would she need to invest, correct to three significant figures ?

$$\textit{principal} = \frac{\textit{interest amount} \times 100 \times 12}{\textit{rate}} = \frac{2750 \times 100 \times 12}{5.4} \approx \$611\ 111.1111\ldots$$

She would need to invest \$612 000 to achieve this payment target.
(Note that investing \$611 000 would give her a monthly payment of \$2749.50 !!)

(f) For an investment of \$550 000, an annuity with an interest rate of 5.39% p.a. will give a greater monthly payment than a perpetuity with an interest rate of 5.4% p.a. Explain why this is so.

An annuity makes a payment of interest plus part of the invested money so that after the agreed period there is no money left. A perpetuity pays only the interest earned, and this can continue indefinitely.

SESSION THREE

PROBLEM ONE

(a) Carla has borrowed \$75 000 to invest in shares. She has taken out an interest-only loan with interest charged at 9.25 % per annum, compounded monthly. She will make monthly interest-only payments and then pay back the full \$75 000 after three years.
Calculate the amount of the monthly payment, correct to the nearest cent.

$$\textit{Interest (payment)} = \frac{9.25 \times 75000}{100 \times 12} = 578.125\ldots \approx \$578.13$$

(b) If she makes \$95 000 from the sale of her shares after three years, calculate how much profit or loss she makes overall on the deal.

$$\textit{Total expenses} = 36 \times \$578.13 + \$75\,000 = \$95\,812.68$$

$$\textit{Loss} = \textit{Expenses} - \textit{Income} = \$95\,812.68 - \$95\,000 = \$812.68$$

(c) Carla wants to borrow money to finance another share trade opportunity. If she can access an interest-only loan at 9.5 % per annum interest, how much could she apply for (to the nearest thousand dollars), if she can afford a **maximum** monthly payment of \$650?

$$\textit{principal} = \frac{\textit{interest amount} \times 100 \times 12}{\textit{rate}} = \frac{650 \times 100 \times 12}{9.5} \approx \$82\,231.5789\ldots$$

She can apply for \$82 000 to keep the monthly repayment UNDER \$650 !!

(d) What is the effective rate of interest, correct to three decimal places, that Carla will be paying on her interest-only loan at 9.5 % per annum, compounded monthly?

$$r_{effective} = \left(\left(1 + \frac{9.5}{100 \times 12}\right)^{12} - 1\right) \times 100\% = 9.924758\ldots \approx 9.925\ \%$$

Use calc eff(12,9.5) = 9.925%

PROBLEM TWO

(a) Diana wants to establish a scholarship fund to support students at her old secondary college. She plans to invest \$50 000 in a term deposit account that pays 6.5 % per annum interest. If the interest is calculated and paid annually as the scholarship amount, how much money will be available each year to be awarded ?

$$\textit{Interest (payment)} = \frac{6.5 \times 50000}{100} = \$3250.00$$

(b) Diana actually wants to have at least \$4000 available each year. How much money will she have to invest, to the nearest thousand dollars, at 6.5 % per annum to earn the required sum each year ?

$$\textit{principal} = \frac{\textit{interest amount} \times 100}{\textit{rate}} = \frac{4000 \times 100}{6.5} \approx \$61\ 538.4615\ldots$$

She will have to invest \$62 000 to meet the interest requirement.
(Note : if she invests \$61 000, she will earn less than \$4000 per year – \$3965 !)

PROBLEM THREE

Naomi runs a dance school and has decided to set up a superannuation fund to look after her in her retirement. She has started the fund with \$8 000 she has saved and invested it in an account earning 5.4 % per annum, compounded monthly. She has decided she can afford to add \$400 each month to the fund.

(a) Use your calculator to determine recursively the value of the superannuation fund, correct to the nearest cent each month, for the first six months, and complete the table below.

Month number	Calculation details	Value of fund
0		8000.00
1	*8 000 × 1.0045 + 400*	*8436.00*
2	*8436.00 × 1.0045 + 400*	*(8873.962…) 8873.96*
3	*8873.962… × 1.0045 + 400*	*(9313.8948…) 9313.89*
4	*9313.8948… × 1.0045 + 400*	*(9755.8073…) 9755.81*
5	*9755.8073… × 1.0045 + 400*	*(10 199.7084…) 10 199.71*
6	*10 199.7084… × 1.0045 + 400*	*(10 645.6071…) 10 645.61*

(b) If she can make monthly payments of \$400 for the next thirty-five years, how much will be in her fund, correct to the nearest cent, when she retires?

Show details of any Finance Solver calculations in the table provided below.

N	**I(%)**	**PV**	**Pmt**	**FV**	**PpY, CpY**
420	*5.4*	*–8 000*	*– 400*	*0*	*12*

*Solve for **FV** : 549 740.986…* *She will have \$549 740.99.*

(c) Naomi believes that she can make a success of her business and she will be able to add more money to her superannuation fund over time. She plans to pay $400 per month for the first ten years, $500 per month for the next ten years, $600 per month for the next ten years, and $700 per month for the last five years. If she continues to receive 5.4 % per annum, how much money will be in her superannuation fund after these thirty-five years?

Show details of any Finance Solver calculations in the table provided below.

N	**I(%)**	**PV**	**Pmt**	**FV**	**PpY, CpY**
120	*5.4*	*– 8 000*	*– 400*	*0*	*12*

*Solve for **FV** : 77 171.8269…* *She will have $77 171.83 after ten years.*

Show details of any Finance Solver calculations in the table provided below.

N	**I(%)**	**PV**	**Pmt**	**FV**	**PpY, CpY**
120	*5.4*	*– 77 171.83*	*– 500*	*0*	*12*

*Solve for **FV** : 211 592.558…* *She will have $211 592.56 after twenty years.*

Show details of any Finance Solver calculations in the table provided below.

N	**I(%)**	**PV**	**Pmt**	**FV**	**PpY, CpY**
120	*5.4*	*– 211 592.56*	*– 600*	*0*	*12*

*Solve for **FV** : 457 845.298…* *She will have $457 845.30 after thirty years.*

Show details of any Finance Solver calculations in the table provided below.

N	**I(%)**	**PV**	**Pmt**	**FV**	**PpY, CpY**
60	*5.4*	*– 457 845.30*	*– 700*	*0*	*12*

*Solve for **FV** : 647 491.2196* *She will have $647 491.22 after thirty-five years.*

PROBLEM FOUR

Henry now works from home as a freelance business consultant after a number of years working in large organisations. He currently has \$200 000 in his superannuation account, but his accountant advises him that he will need to have at least \$750 000 in the account when he plans to retire in fifteen years.

If his superannuation account is earning 5.75 % interest per annum, compounded monthly, how much will Henry have to add each month, rounded to the nearest dollar, to at least meet his intended goal in fifteen years?

Show details of any Finance Solver calculations in the table provided below.

N	I(%)	PV	Pmt	FV	PpY, CpY
180	*5.75*	*– 200 000*	*0*	*750 000*	*12*

*Solve for **Pmt** : – 973.5054…*
He will have to contribute \$974 per month (which will give him \$750 140)